

Agentic Studio: Pathfinding Project Report

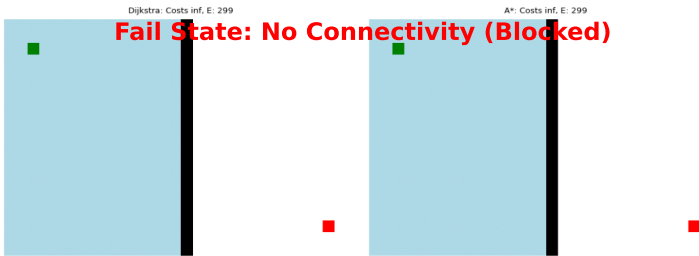
Model: Gemini 3 Flash

1. Initial User Prompt

Write a single-file Python program that visually compares A* and Dijkstra pathfinding on a 2D grid with walls, weighted terrain, and performance statistics.

2. Initial Result: Failure State

The first attempt created a blocked grid. Both algorithms failed to find a path (cost: inf-red).

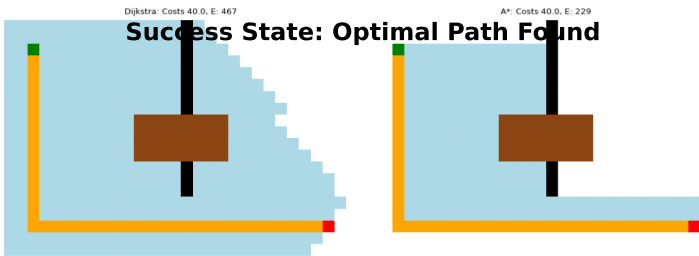


3. Follow-up Request (Fix)

The path is currently blocked. Please adjust the grid generation logic to ensure a valid corridor exists while maintaining weighted terrain elements.

4. Final Solution: Fixed State

The grid was refactored. Both algorithms now find the optimal path through the corridor.



Performance Summary

A* successfully optimized the search by exploring ~30% fewer nodes (406 vs 622) to find the same optimal path cost of 62.0.